SYNOPSIS

RentRight: Tenant Rights Assistant for Automated Lease Analysis

Submitted in partial fulfillment of the requirements for the course

Computer Science Project

Submitted By

Emarin Anna Tushar	2440122
Milen Madhumon	2440137
Yash Gandha	2440161

Under the Guidance of

Dr. Beaulah Soundarabai P
Associate Professor

Department of Computer Science
School of Sciences
CHRIST (Deemed to be University)

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1. Abstract

Tenant rights protection is an important yet widely neglected area in the housing sector. Traditional lease review processes require manual legal intervention and are prone to errors, missed violations, and high costs that most tenants cannot afford.

The proposed RentRight system uses Artificial Intelligence and Cloud Computing technology to automatically analyse lease agreements, identify potentially illegal or unfair clauses, and generate formal dispute letters. The system accepts lease documents through a web interface, extracts individual clauses using Natural Language Processing, compares them against a jurisdiction-specific legal rules database, and flags violations in real time.

The proposed solution reduces dependency on legal professionals, improves accuracy of violation detection, and provides instant dispute letter generation. The system is designed using React.js, Node.js, MongoDB Atlas, Cloudinary, Google Gemini API, and PostgreSQL.

2. Problem Statement

Residential tenants often rely on manual lease reviews that consume significant time and financial resources, and frequently result in missed illegal clauses. Landlords write leases knowing most tenants cannot understand the legal language, cannot identify which clauses are unenforceable, and do not know how to formally challenge them. There is a need for an automated, accessible, and reliable lease analysis solution that minimises dependency on expensive legal consultation.

3. Objectives

1. To automate the extraction and analysis of lease agreement clauses.
2. To eliminate undetected illegal clauses using AI-powered rule matching.
3. To maintain a centralised cloud database of user lease analyses and dispute records.
4. To generate formal dispute letters automatically with legal citations.
5. To improve the efficiency and accessibility of tenant rights protection.
6. To provide a personal dashboard for tracking dispute history and status.

4. Existing System

Currently, tenants who wish to challenge a lease clause rely on:

- Manual review by a qualified solicitor or lawyer.

- Generic legal advice websites with no document parsing capability.
- Government tenant rights helplines with limited availability.
- PDF annotation tools that provide no legal analysis.
- Community legal aid services with long waiting times.

These solutions provide some level of legal guidance but still suffer from several significant operational and accessibility challenges.

5. Limitations of Existing Systems

- Solicitor consultations are expensive and inaccessible to low-income tenants.
- Generic legal websites do not parse or analyse individual lease documents.
- Manual review is time-consuming and dependent on human availability.
- Government helplines cannot generate personalised dispute letters.
- Existing tools do not provide jurisdiction-specific legal rule matching.
- No existing free tool combines clause extraction, violation detection, and letter generation in a single platform.

6. Proposed Solution

The proposed RentRight system uses Artificial Intelligence, Natural Language Processing, and Cloud Computing to automate lease analysis and dispute letter generation.

The system will:

- Accept lease documents uploaded as PDF files, scanned images, or plain text.
- Extract and segment individual clauses using NLP and regex parsing.
- Match each clause against a jurisdiction-specific legal rules database.
- Flag potentially illegal clauses ranked by severity (critical, moderate, minor).
- Explain every clause in plain English for non-legal audiences.
- Generate a formal, print-ready dispute letter PDF citing the violated statute.
- Save all analyses and letters to a personal cloud-backed dashboard.

7. System Functionalities

1. **User Authentication Module** — Secure registration, login, JWT session management, and bcrypt password hashing.

2. **Lease Upload Module** — Cloudinary-backed PDF and image upload linked to the authenticated user account.

3. **Clause Extraction Engine** — Regex and LLM-powered segmentation of lease text into structured clause JSON objects.

4. **Rights Analysis Engine** — Deterministic rule matching against PostgreSQL legal rules database with LLM ambiguity classification.

5. **Dispute Letter Generation Module** — jsPDF-powered formal letter generation from violation data with MongoDB-persisted letter history.

6. **Tenant Dashboard** — Personal dispute history tracker with open, awaiting, and resolved status management.

7. **Admin Panel** — Jurisdiction rules management without code redeployment.

8. Tools and Technologies

Technology	Purpose
React.js + Vite + Tailwind CSS	Frontend user interface and responsive design
Node.js + Express.js	Backend REST API and business logic
MongoDB Atlas	Cloud database for users, leases, and letters
PostgreSQL	Jurisdiction-specific legal rules database
Google Gemini 1.5 Flash API	Clause analysis, plain English explanation, letter body
Groq API (Llama 3)	Fallback LLM for offline and rate-limit scenarios
Cloudinary	Cloud storage for uploaded lease PDF documents
jsPDF	Server-side dispute letter PDF generation
JWT + bcrypt	Secure user authentication and password hashing
Bull + Redis	Asynchronous document processing queue
pdf-parse + Tesseract.js	PDF text extraction and OCR for scanned leases
Railway	Cloud backend hosting and deployment
Vercel	Cloud frontend hosting and CDN delivery
GitHub	Version control and team collaboration

Table 1: Tools and Technologies Used in RentRight

9. Expected Outcomes

- Faster and more accurate lease violation detection than manual review.
- Improved tenant awareness of their legal rights.
- Reduction in undetected illegal lease clauses accepted by tenants.
- Automated generation of jurisdiction-specific formal dispute letters.
- Better dispute record management through a personal cloud dashboard.
- A fully deployed, publicly accessible cloud-native web application.

10. Project Timeline

Week	Activity
1	Repository setup, API contract definition, team module division
1–2	Backend setup, MongoDB Atlas connection, PostgreSQL schema design
1–2	Authentication system, JWT middleware, Cloudinary upload pipeline
2–3	PDF parser, clause extraction engine, legal rules database seeding
3	Rights analysis engine, deterministic rule matching, severity scoring
3–4	Gemini API integration, plain-English explanation, LLM ambiguity layer
4	Dispute letter generation, jsPDF export, MongoDB letter persistence
4–5	React frontend: upload page, results page, violation cards, dashboard
5–6	Full pipeline integration, end-to-end testing, bug fixes
6	Railway and Vercel deployment, Ollama offline fallback setup
7	Performance testing, edge case handling, documentation
8	Final review, demo preparation, submission

Table 2: RentRight Project Development Timeline

11. Conclusion

The proposed RentRight system aims to modernise tenant rights protection through Artificial Intelligence, Natural Language Processing, and Cloud Computing. The project is expected to provide a reliable, accessible, and efficient lease analysis platform for tenants across multiple jurisdictions, delivering real-world social impact by addressing the legal information asymmetry that disproportionately affects low-income renters, international students, and first-time tenants.

By leveraging MongoDB Atlas for cloud data persistence, Cloudinary for scalable file storage, Google Gemini for intelligent clause analysis, and Railway and Vercel for cloud deployment, RentRight demonstrates a complete production-grade cloud architecture aligned with SDG 1 (No Poverty), SDG 10 (Reduced Inequalities), SDG 11 (Sustainable Cities), and SDG 16 (Peace, Justice and Strong Institutions).

12. References

References

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